



Indian Institute of Technology Ropar
Department of Mathematics
MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 8

1. Consider the following Transportation Problem

	40	50	10
40	2	5	2
30	4	1	3
30	3	6	2

Determine a starting basic feasible solution (b.f.s) by employing:

- (a) the north-west corner rule,
 - (b) the column minima,
 - (c) the row minima,
 - (d) the matrix minima, and
 - (e) VAM (Vogel's Approximation Method).
2. Find an optimal solution to the following transportation problem.

	D_1	D_2	D_3	D_4	Supply (Availability)
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
<i>Demand (Requirement)</i>	5	8	7	14	34

(Use Vogel's approx. method to find IBFS).

3. Find an optimal solution to the following Unbalanced transportation problem.

	D_1	D_2	D_3	Supply (Availability)
S_1	4	8	8	76
S_2	16	24	16	82
S_3	8	16	24	77
<i>Demand (Requirement)</i>	72	102	41	

(Use the north-west corner rule to find IBFS).

4. A company has four warehouses and six stores. The total supply available at the warehouses is 22 units, and the total demand at the stores is also 22 units.

Warehouses	Stores						Supply
	1	2	3	4	5	6	
1	9	12	9	6	9	10	5
2	7	3	7	7	5	5	6
3	6	5	9	11	3	11	2
4	6	8	11	2	2	10	9
Demand	4	4	6	2	4	2	

- (a) Obtain an initial basic feasible solution using Vogel's Approximation Method (VAM).
- (b) Check whether the solution is degenerate.
- (c) If degeneracy occurs, resolve it using an appropriate technique.
- (d) Obtain the optimal solution.
5. If $\{x_{13}, x_{22}, x_{23}, x_{31}, x_{32}, x_{34}\}$ (with $x_{23} = 10$) is the set of basic cells of a balanced transportation problem seeking to minimize the cost of transportation from origins to destination, where the cost matrix is,

	D_1	D_2	D_3	D_4	Availability
O_1	6	2	-1	0	10
O_2	4	2	2	3	$\lambda+5$
O_3	3	1	2	1	3λ
Demand	10	$\mu - 5$	$\mu + 5$	15	

and $\lambda, \mu \in \mathbb{R}$. Then find values of λ, μ and $x_{32} = \underline{\hspace{1cm}}$.